

Variation of Parameters

ANSWER KEY



The Wronskian

- 1 For $y_1 = e^x$, $y_2 = e^{2x}$ (solutions of $y'' - 3y' + 2y = 0$), compute the Wronskian $W = y_1 y_2' - y_1' y_2$.
 $y_1' = e^x$, $y_2' = 2e^{2x}$. $W = e^x(2e^{2x}) - e^x(e^{2x}) = 2e^{3x} - e^{3x} = e^{3x}$.
- 2 For $y_1 = \cos x$, $y_2 = \sin x$, compute W .
 $y_1' = -\sin x$, $y_2' = \cos x$. $W = \cos x \cos x - (-\sin x) \sin x = \cos^2 x + \sin^2 x = 1$.

The variation of parameters formula

- 3 State the variation of parameters formula for a particular solution of $y'' + p(x)y' + q(x)y = g(x)$, given independent solutions y_1, y_2 of the homogeneous equation with Wronskian W .
 $y_p = -y_1 \int \frac{y_2 g}{W} dx + y_2 \int \frac{y_1 g}{W} dx$.

A forcing term with no undetermined-coefficients trial form

- 4 Solve $y'' + y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.
Here $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$, $g = \tan x$. Since $\int \cos x \tan x dx = \int \sin x dx = -\cos x$ and $\int \sin x \tan x dx = \int \sec x dx - \int \cos x dx = \ln|\sec x + \tan x| - \sin x$, we get
 $y_p = -\cos x[\ln|\sec x + \tan x| - \sin x] + \sin x(-\cos x) = -\cos x \ln|\sec x + \tan x|$. General solution:
 $y = C_1 \cos x + C_2 \sin x - \cos x \ln|\sec x + \tan x|$.

A repeated root, with forcing dividing by x

- 5 Solve $y'' - 2y' + y = \frac{e^x}{x}$ ($x > 0$) using variation of parameters.
 $r^2 - 2r + 1 = (r - 1)^2 = 0$, so $y_1 = e^x$, $y_2 = xe^x$, and $W = e^{2x}$. With $g = e^x/x$: $\frac{y_2 g}{W} = 1$ so $\int = x$; $\frac{y_1 g}{W} = \frac{1}{x}$ so $\int = \ln x$. Then $y_p = -e^x(x) + xe^x(\ln x) = xe^x(\ln x - 1)$. Absorbing the $-xe^x$ term into the arbitrary constant on y_2 , the general solution is $y = C_1 e^x + C_2 x e^x + x e^x \ln x$.

Comparing with undetermined coefficients

- 6 Solve $y'' - y = e^{2x}$ by variation of parameters, then check the particular solution using undetermined coefficients.
 $y_1 = e^x$, $y_2 = e^{-x}$, $W = -2$. With $g = e^{2x}$: $\frac{y_2 g}{W} = -\frac{1}{2}e^x$, integral $-\frac{1}{2}e^x$; $\frac{y_1 g}{W} = -\frac{1}{2}e^{3x}$, integral $-\frac{1}{6}e^{3x}$. So $y_p = -e^x(-\frac{1}{2}e^x) + e^{-x}(-\frac{1}{6}e^{3x}) = \frac{1}{2}e^{2x} - \frac{1}{6}e^{2x} = \frac{1}{3}e^{2x}$. Check by undetermined coefficients: try Ae^{2x} (no resonance, since 2 is not a root of $r^2 - 1 = 0$): $4A - A = 1 \Rightarrow A = \frac{1}{3}$. Matches. General solution:
 $y = C_1 e^x + C_2 e^{-x} + \frac{1}{3}e^{2x}$.